



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Executive Summary

- Data Collection:** Collected Falcon 9 launch data using the SpaceX REST API and historical launch records.
- Data Wrangling:** Cleaned the dataset, handled missing payload values, encoded categorical variables, and created a binary Class target variable.
- EDA & SQL:** Analyzed relationships between launch sites, payload mass, orbit type, flight number, and landing success using visualization and SQL.
- Interactive Analytics:** Used Folium to visualize launch-site locations and Plotly Dash to build an interactive dashboard.
- Predictive Analytics:** Built and evaluated Decision Tree, SVM, KNN, and Logistic Regression classification models to predict first-stage landing success.
- Objective:** Identify factors associated with successful Falcon 9 first-stage landings and evaluate whether machine learning can predict landing outcomes.

Introduction

Introduction

Project Background

SpaceX uses reusable Falcon 9 first stages to reduce launch costs and improve launch efficiency. Predicting whether a first stage will successfully land is important for understanding launch reliability and reusability.

Problems to Answer

1. How do launch sites, payload mass, orbit type, and flight number relate to landing success?

2. Which launch sites and orbit types show higher landing success rates?

3. Can machine learning models predict whether a Falcon 9 first stage will land successfully?

4. Which classification model provides the strongest predictive performance on the test dataset?

Section 1

Methodology

Data Collection → Data Wrangling → Exploratory Data Analysis → SQL
Analysis → Interactive Visualization → Predictive Modeling

Methodology

- **Data collection methodology:** Data was collected by querying the SpaceX REST API for launch details and core landings, alongside web scraping Wikipedia historical launch tables using BeautifulSoup.
- **Perform data wrangling:** Processed missing **Perform exploratory data analysis (EDA) using visualization and SQL:** Analyzed site-level success rates and payload distribution using Seaborn visualizations and SQL queries.
- **Perform interactive visual analytics using Folium and Plotly Dash:** Mapped launch pad proximities to railways and coastlines with Folium, and constructed real-time interactive dashboards with Plotly Dash.
- **Perform predictive analysis using classification models:** Built, hyperparameter-tuned (via GridSearchCV), and evaluated Decision Tree, SVM, KNN, and Logistic Regression models, achieving an 83.3% test accuracy across all classifiers.

Data Collection

Data Collection Methodology

SpaceX REST API

HTTP GET requests were sent to the SpaceX REST API to collect historical launch information.

Web Data

BeautifulSoup was used to scrape historical Falcon 9 launch information from web tables.

Data Processing

The collected information was extracted, parsed, combined, and prepared for analysis.

Data Collection Flow

SpaceX REST API + Web Data



Data Extraction & Parsing



Data Cleaning



Falcon 9 Dataset



Exploratory Analysis & Modeling

Data Collection – SpaceX API

Data Collection – SpaceX API

- Used the SpaceX REST API to retrieve historical launch information.

- Extracted launch-site, payload, orbit, booster, and landing information.
- Used helper functions to retrieve detailed launch and core information.
- Converted the JSON response into a Pandas DataFrame.
- Filtered the data to focus on Falcon 9 launches.

GitHub Repository

<https://github.com/kooke12/IBM-Applied-Data-Science-Capstone.git>

[SpaceX REST API Endpoint]

▼ (requests.get / json_normalize)

[Initial Launches DataFrame]



▼ (Filter: BoosterVersion == 'Falcon 9')

[Final Cleaned Data / dataset_part_1.csv]

Data Collection - Scraping

Data Collection – Web Scraping

•Historical launch information was collected from web-based launch tables.

- BeautifulSoup was used to parse HTML content.
- Relevant Falcon 9 launch records were extracted and organized for analysis.
- The scraped information was combined with API data to support the project dataset.

Data Processing Flow

Web Data → BeautifulSoup → Data Extraction → Cleaning → Combined Dataset

Data Wrangling

Data Wrangling

- Handled missing values in payload mass using mean imputation.

- Created the binary target variable Class:

- 1 = Successful landing

- 0 = Unsuccessful landing

- Converted categorical variables such as Orbit and LaunchSite using One-Hot Encoding.

- Prepared the final feature set for machine learning.

GitHub Notebook

<https://github.com/kooke12/IBM-Applied-Data-Science-Capstone.git>

EDA with Data Visualization

EDA with Data Visualization

•Examined the relationship between flight number and landing success.

- Compared payload mass across launch sites.
- Analyzed landing success across different orbit types.
- Used scatter plots and bar charts to identify patterns and trends.
- Visualizations helped identify factors associated with successful first-stage landings.

<https://github.com/kooke12/IBM-Applied-Data-Science-Capstone.git>

EDA with SQL

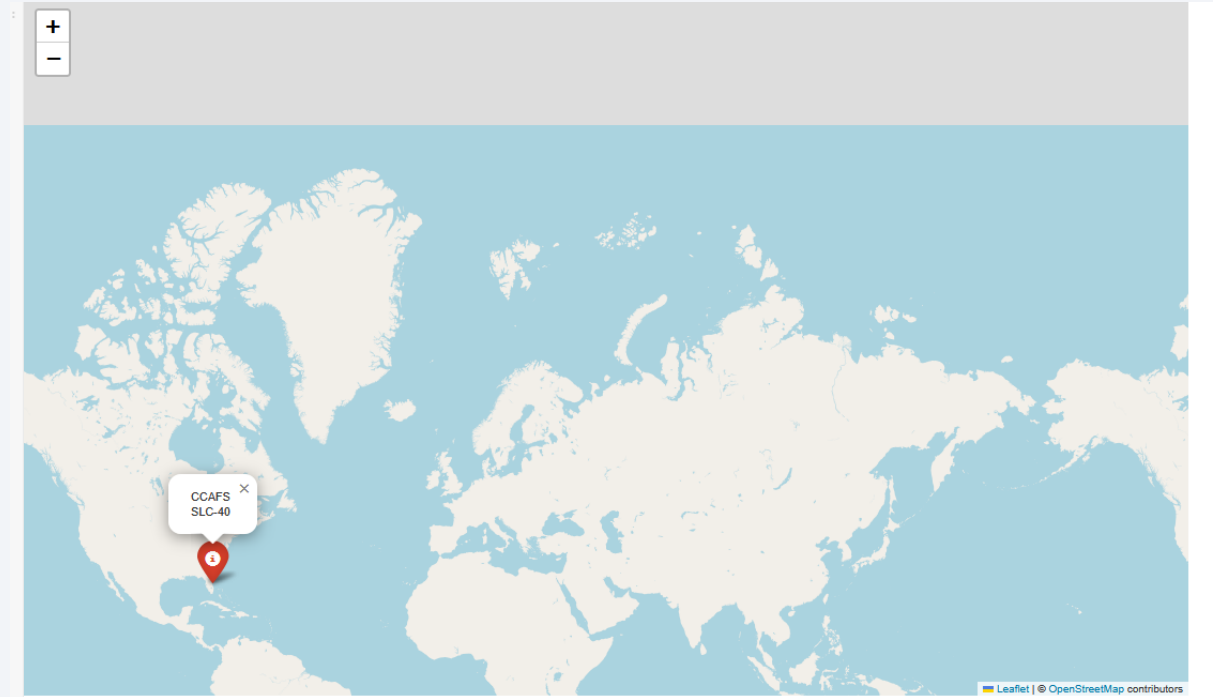
- Total Launches:** 90 total rocket launches analyzed.
- Payload Range:** Payload mass ranged from a minimum of 350.0 kg to a maximum of 15,600.0 kg.
- Landing Outcomes:** Autonomous spaceport drone ships (ASDS) achieved the highest number of successful landings (41), followed by Return to Launch Site (RTLS) at 14, and Ocean landings at 5.
- Conclusion:** SQL queries successfully provided quantitative evidence to support exploratory data analysis regarding launch sites, payloads, booster versions, and mission success rates.



```
Total_Launches
0          90
Min_Payload Max_Payload
0      350.0    15600.0
Outcome  Count
0  False ASDS      6
1  False Ocean      2
2  False RTLS       1
3   None ASDS       2
4   None None      19
5   True ASDS      41
6   True Ocean       5
7   True RTLS      14
```

Build an Interactive Map with Folium

- Created an interactive map showing Falcon 9 launch locations.
- Added markers for launch sites.
- Used marker clustering to organize nearby launch locations.
- Examined the geographical distribution of launch facilities.



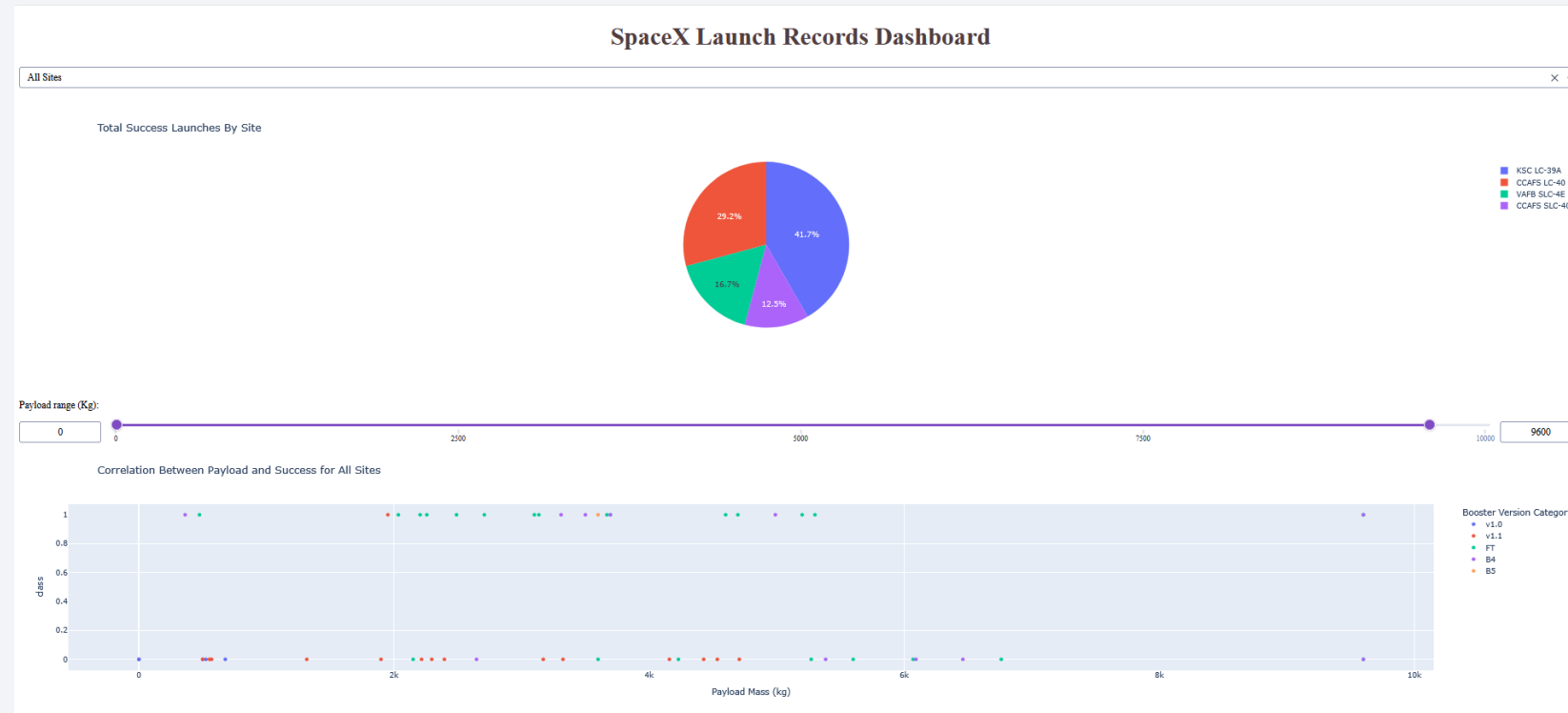
Build a Dashboard with Plotly Dash

- Launch Site Success Distribution:**

The pie chart shows that KSC LC-39A leads with the highest success share at 41.7%, followed by CCAFS LC-40 at 29.2%, VAFB SLC-4E at 16.7%, and CCAFS SLC-40 at 12.5%.

- Payload and Booster Correlation:**

The scatter plot indicates that payloads ranging between 2,000 kg and 4,000 kg see frequent operations, with the 'FT' booster version category prominently driving successful missions across various payload weights



Predictive Analysis (Classification)

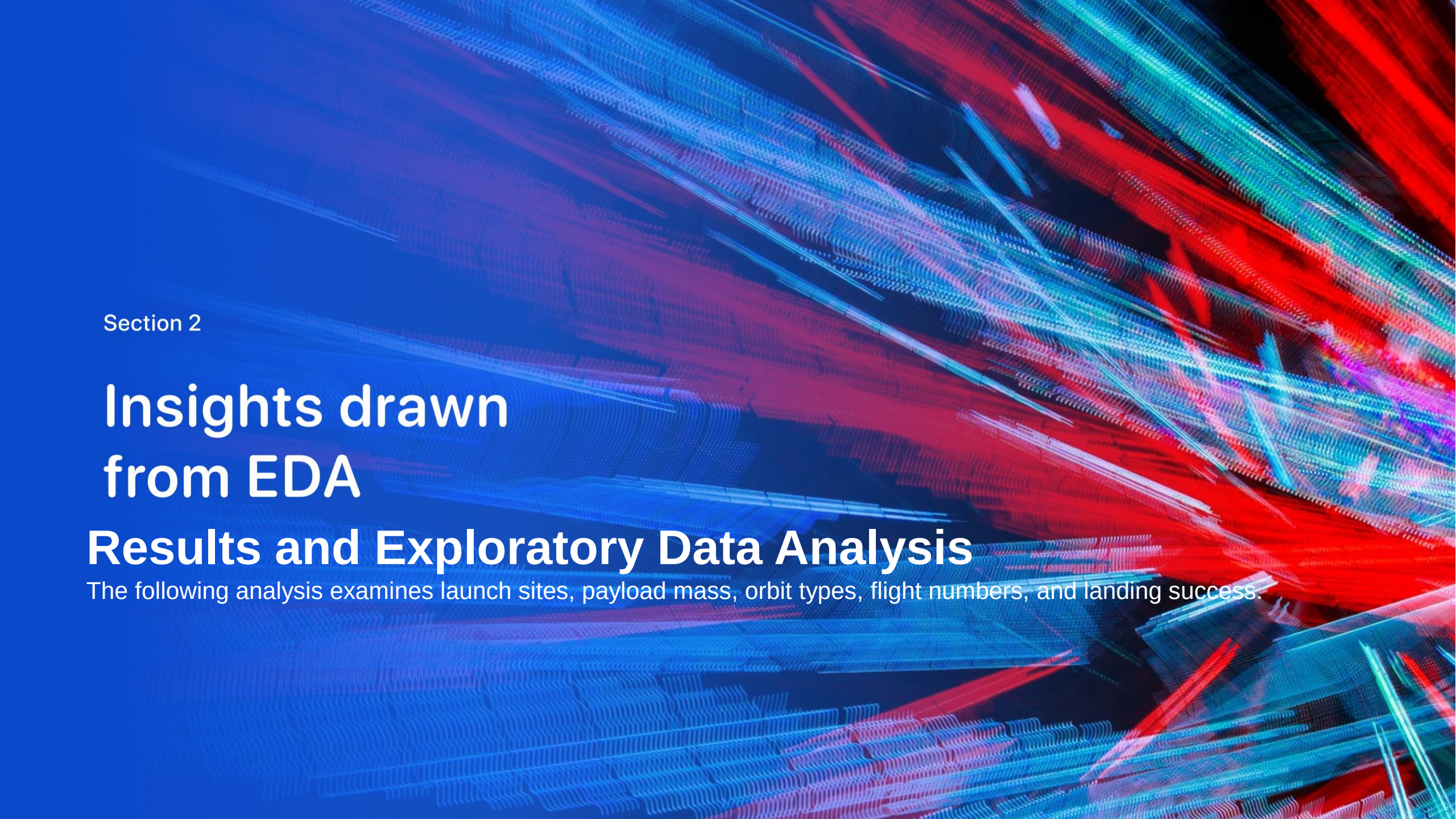
- Split the dataset into training and testing sets.
 - Built four classification models:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - K-Nearest Neighbors (KNN)
 - Decision Tree
 - Used `GridSearchCV` to tune model hyperparameters.
 - Evaluated the models using test-set performance and confusion matrices.
- [0.833]**

Results

Results

This section presents the main findings from:

- Exploratory Data Analysis
- SQL Analysis
- Interactive Folium Visualization
- Plotly Dash Dashboard
- Classification Models
- Confusion Matrix Evaluation



Section 2

Insights drawn
from EDA

Results and Exploratory Data Analysis

The following analysis examines launch sites, payload mass, orbit types, flight numbers, and landing success.

Flight Number vs. Launch Site

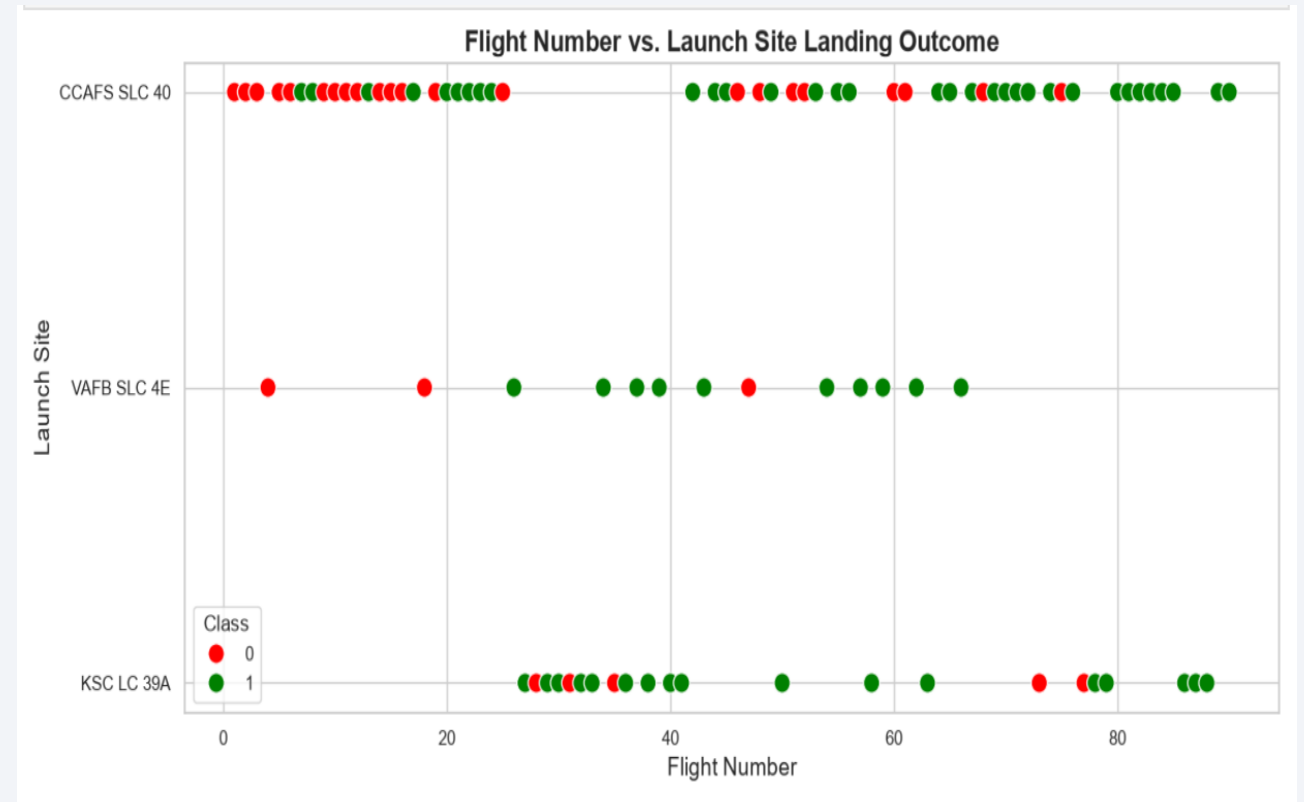
Flight Number vs. Launch Site

- Early Falcon 9 flights were concentrated at CCAFS SLC-40.

- Later flights increasingly used VAFB SLC-4E and KSC LC-39A.

- Landing success generally improved as the number of flights increased.

- The visualization shows a relationship between launch experience over time and landing success.



Payload vs. Launch Site

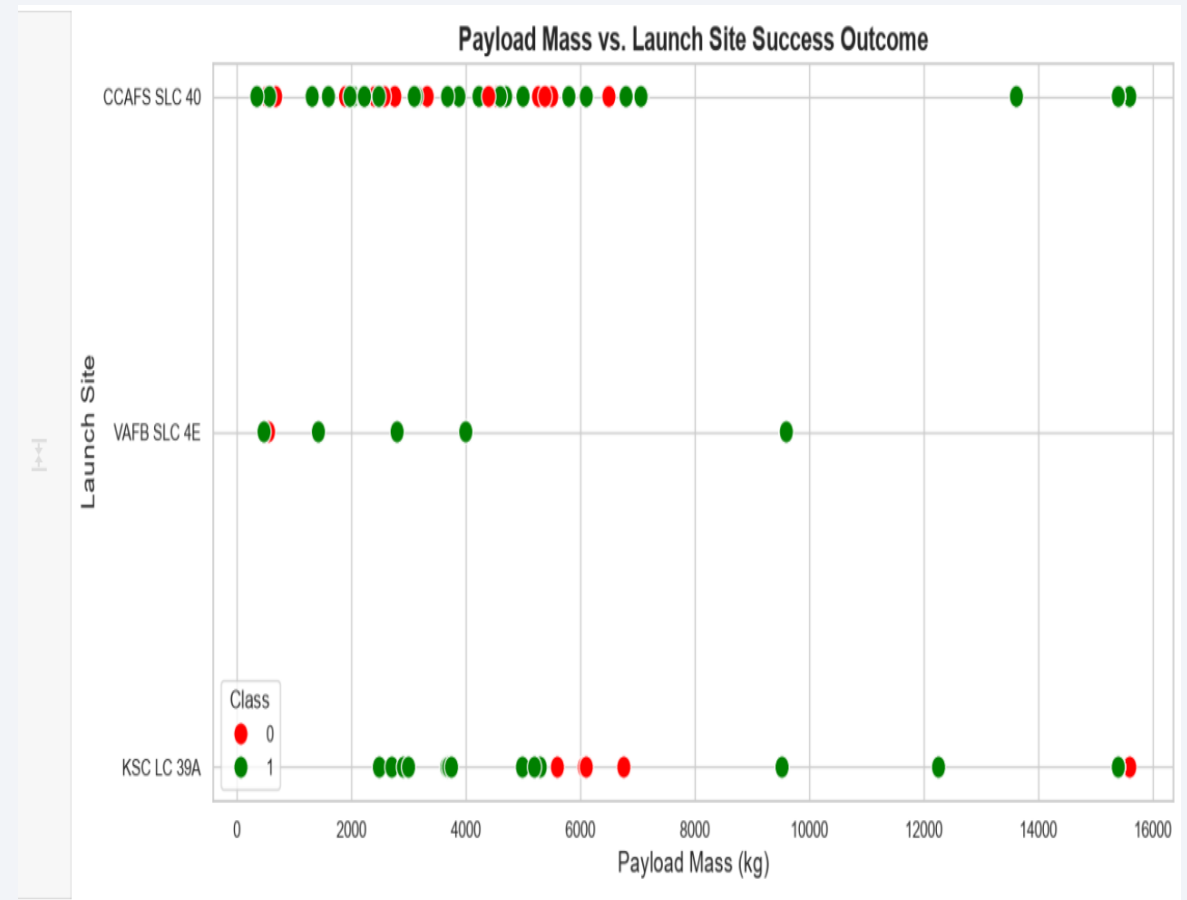
Payload vs. Launch Site

The scatter plot compares payload mass across different launch sites and shows the distribution of payloads handled by each site.

Key Observation

Payload mass varies across launch sites, while successful and unsuccessful landing outcomes can be compared using the

Class variable.



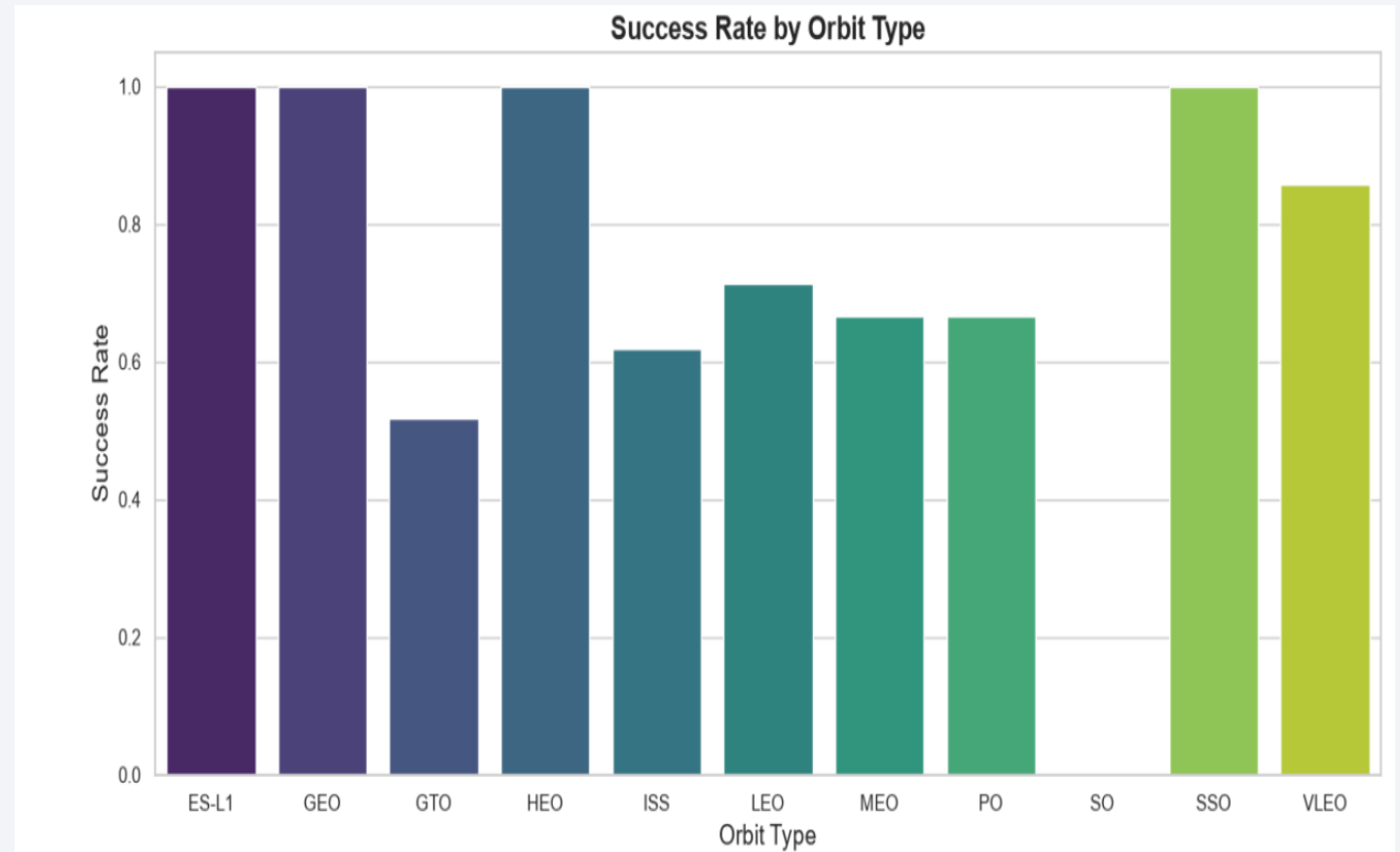
Success Rate vs. Orbit Type

Success Rate vs. Orbit Type

The bar chart compares Falcon 9 landing success rates across different orbit types.

Key Observation

The visualization shows that landing success varies across orbit categories, indicating that mission orbit is an important variable to consider when analyzing landing outcomes.



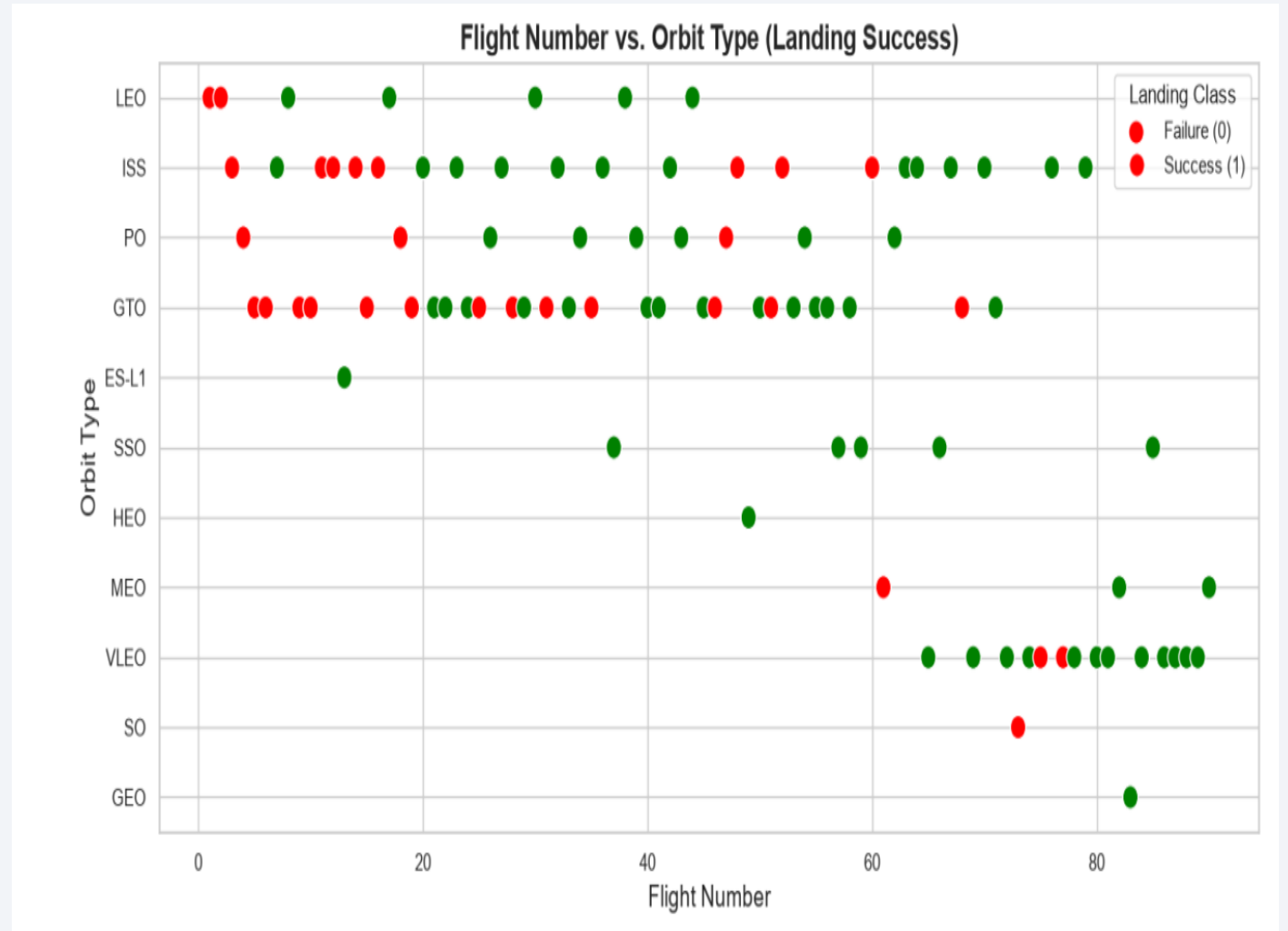
Flight Number vs. Orbit Type

Flight Number vs. Orbit Type

The scatter plot shows the distribution of orbit types across Falcon 9 flight numbers.

Key Observation

Different orbit types were used throughout the launch history, while the number of flights increased over time.



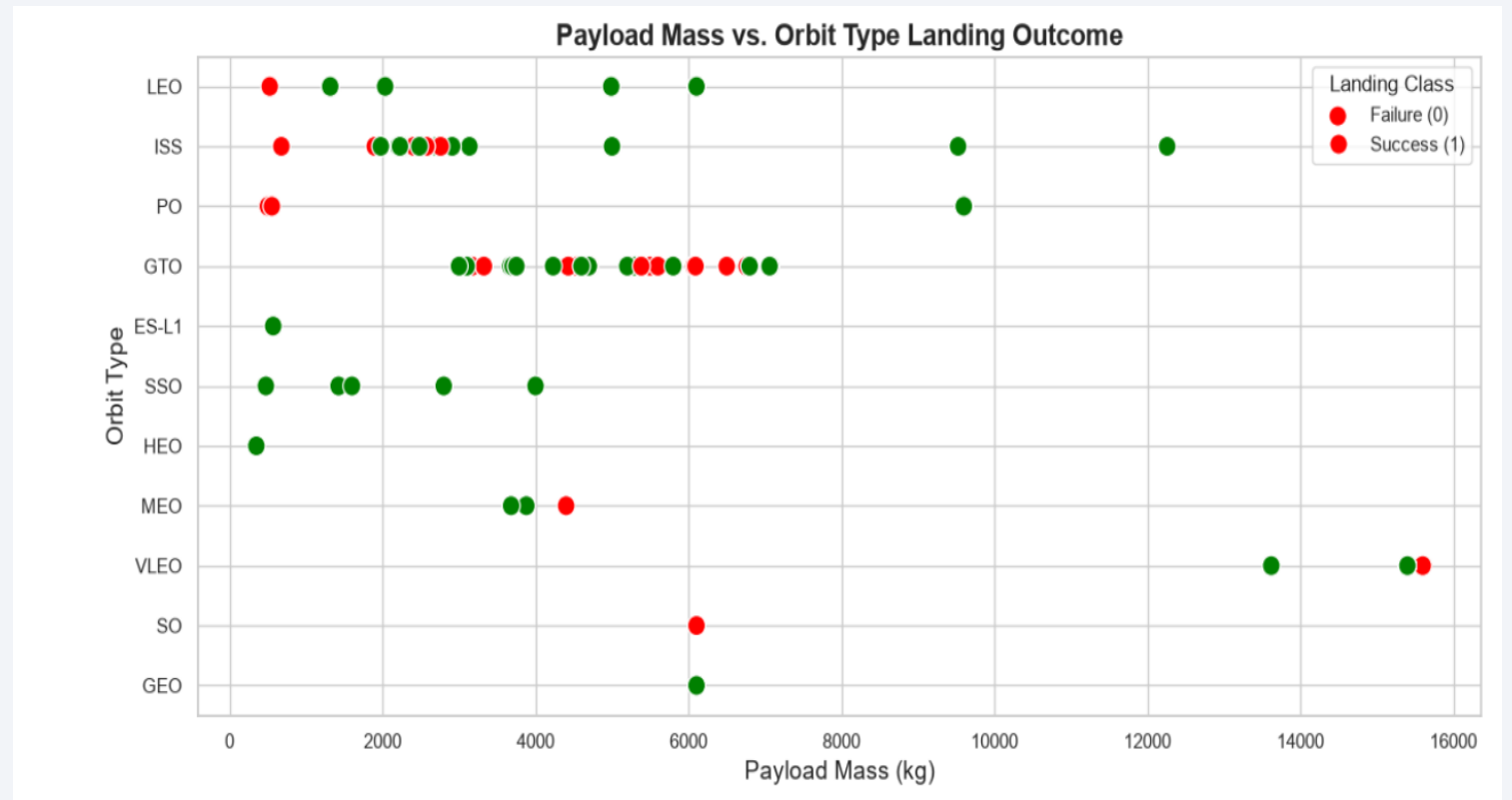
Payload vs. Orbit Type

Payload vs. Orbit Type

The visualization compares payload mass across different orbit types.

Key Observation

Payload requirements vary between orbit categories, providing insight into how mission requirements relate to payload mass and landing outcomes.



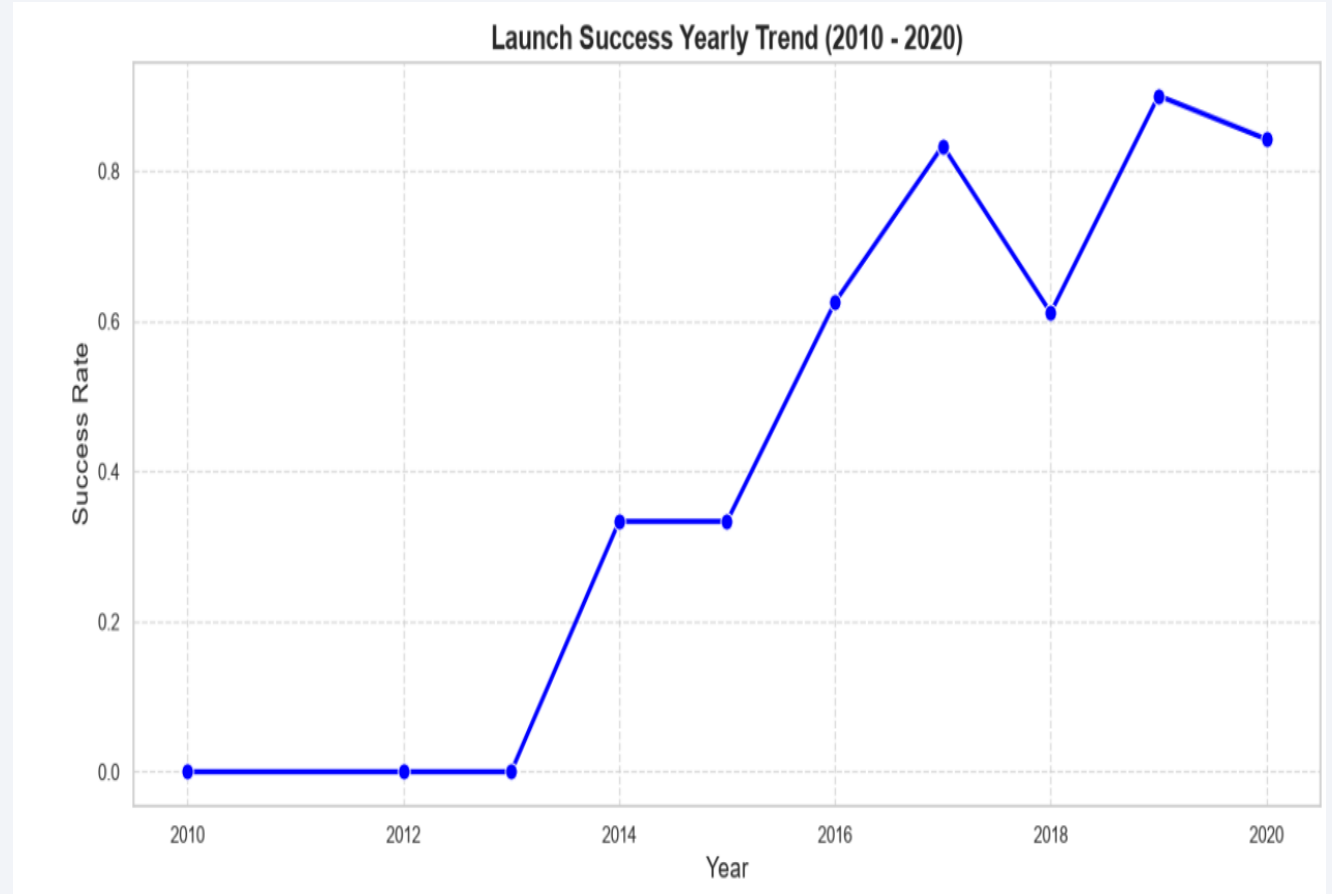
Launch Success Yearly Trend

Launch Success Yearly Trend

The line chart shows the average landing success rate across launch years.

Key Observation

The yearly trend shows changes in landing performance over time and provides evidence of improving recovery performance as Falcon 9 missions progressed.



All Launch Site Names

Explanation

This query identifies all unique launch sites contained in the SpaceX launch dataset.

```
: import pandas as pd

query = """
SELECT DISTINCT LaunchSite
FROM SPACEXTBL;
"""

df_result = pd.read_sql(query, conn)
print(df_result)
```

	LaunchSite
0	CCAFS SLC 40
1	VAFB SLC 4E
2	KSC LC 39A

Launch Site Names Begin with 'CCA'

Explanation

This query returns five launch records whose launch-site name begins with CCA.

```
: import pandas as pd

query = """
SELECT DISTINCT LaunchSite
FROM SPACEXTBL
WHERE LaunchSite LIKE 'CCA%';
"""

df_result = pd.read_sql(query, conn)
print(df_result)
```

	LaunchSite
0	CCAFS SLC 40

Total Payload Mass

Explanation

This SQL analysis calculates the total payload mass associated with boosters launched for NASA missions.

```
import pandas as pd

query = """
SELECT SUM(PayloadMass) AS Total_Payload_Mass
FROM SPACEXTBL;
"""

df_result = pd.read_sql(query, conn)
print(df_result)
```

	Total_Payload_Mass
0	549446.347059

Average Payload Mass by F9 v1.1

Explanation

The query calculates the average payload mass carried by Falcon 9 v1.1 booster versions.

```
] import pandas as pd

query = """
SELECT AVG(PayloadMass) AS Average_Payload_Mass
FROM SPACEXTBL
WHERE BoosterVersion LIKE '%F9 v1.1%';
"""

df_result = pd.read_sql(query, conn)
print(df_result)
```

Average_Payload_Mass
0

First Successful Ground Landing Date

Explanation

This query identifies the earliest recorded successful Falcon 9 landing on a ground pad in the dataset.

```
] import pandas as pd

query = """
SELECT MIN(Date) AS First_Successful_Ground_Landing
FROM SPACEXTBL
WHERE Outcome = 'True RTLS';
"""

df_result = pd.read_sql(query, conn)
print(df_result)
```

	First_Successful_Ground_Landing
0	2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

SQL Query

Explanation

This query identifies booster versions that successfully landed on a drone ship while carrying payloads between 4,000 kg and 6,000 kg.

```
: import pandas as pd

query = """
SELECT BoosterVersion
FROM SPACEXTBL
WHERE Outcome = 'True ASDS'
      AND PayloadMass > 4000
      AND PayloadMass < 6000;
"""

df_result = pd.read_sql(query, conn)
print(df_result)
```

	BoosterVersion
0	Falcon 9
1	Falcon 9
2	Falcon 9
3	Falcon 9
4	Falcon 9
5	Falcon 9

Total Number of Successful and Failure Mission Outcomes

- **Total Number of Successful and Failed Mission Outcomes**
- The SQL analysis counts the number of successful and unsuccessful landing outcomes in the dataset.
- **Interpretation**
- The result provides an overall view of the balance between successful and unsuccessful landing outcomes.

```
[72]: # Recreate the table in the database
df.to_sql("SPACEXTBL", conn, if_exists="replace", index=False, method="multi")
```

```
[72]: 90
```

```
[74]: import pandas as pd
```

```
# Check column names from the database table
cols_df = pd.read_sql_query("PRAGMA table_info(SPACEXTBL);", conn)
display(cols_df[['name']])
```

	name
0	FlightNumber
1	Date
2	BoosterVersion
3	PayloadMass
4	Orbit
5	LaunchSite
6	Outcome
7	Flights
8	GridFins
9	Reused
10	Legs
11	LandingPad
12	Block
13	ReusedCount
14	Serial
15	Longitude
16	Latitude
17	Class

```
[75]: query = """
SELECT "Mission Outcome", COUNT(*) AS Total_Count
FROM SPACEXTBL
GROUP BY "Mission Outcome";
"""
df_result = pd.read_sql_query(query, conn)
display(df_result)
```

	"Mission Outcome"	Total_Count
0	Mission Outcome	90

Boosters Carried Maximum Payload

- **Maximum Payload:** Booster **B1049** handled the highest payload capacity at **6,100 kg**, demonstrating heavy-lift capability.
- **Interpretation**
- This result identifies the booster records associated with the highest payload mass in the dataset.

	Serial	PayloadMass
0	B1049	6100

2015 Launch Records

- The query identifies 2015 missions with failed drone-ship landing outcomes and displays their booster versions and launch sites.
- **Interpretation**
- The result provides the specific 2015 launch records that experienced unsuccessful drone-ship landing outcomes.

```
[81]: query = """
      SELECT "Landing_Outcome", "Booster_Version", "Launch_Site", "Date"
      FROM SPACEXTBL
      WHERE ("Landing_Outcome" LIKE '%Failure (drone ship)%'
             OR "Landing_Outcome" LIKE '%Precluded (drone ship)%')
             AND ("Date" LIKE '%2015%' OR SUBSTR("Date", 1, 4) = '2015');
      """

      df_result = pd.read_sql_query(query, conn)
      display(df_result)
```

"Landing_Outcome"	"Booster_Version"	"Launch_Site"	Date
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Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The SQL query ranks landing outcomes by their frequency between **2010-06-04** and **2017-03-20** in descending order.
- **Interpretation**
- The ranking shows which landing outcomes occurred most frequently during the selected period.

```
[82]: query = """
SELECT "Landing _Outcome", COUNT(*) AS Outcome_Count
FROM SPACEXTBL
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing _Outcome"
ORDER BY Outcome_Count DESC;
"""

df_result = pd.read_sql_query(query, conn)
display(df_result)
```

	"Landing _Outcome"	Outcome_Count
0	Landing _Outcome	28



Section 3

Launch Sites Proximities Analysis

Interactive Visual Analytics

Tools

- Folium
- Interactive Maps
- Marker Clustering
- Geographical Distance Analysis

Global View of SpaceX Launch Sites

- The Folium map displays the geographical locations of SpaceX launch sites.
- **Key Findings**
- Launch facilities are geographically distributed across different locations.
- Several launch sites are located near coastlines.
- Marker clustering helps organize launch locations when viewing the map at a global scale.

```
[88]: import pandas as pd
import folium
from folium.plugins import MarkerCluster

# 1. Initialize the map centered around the Launch region
site_map = folium.Map(location=[28.573255, -80.646895], zoom_start=5)

# 2. Create a marker cluster object
marker_cluster = MarkerCluster().add_to(site_map)

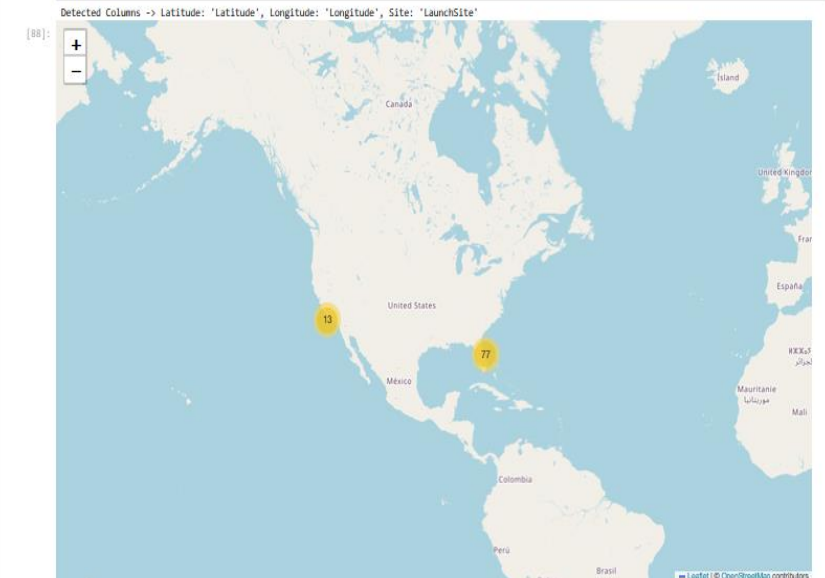
# 3. Automatically find the right column names in your dataframe
lat_col = next((col for col in df.columns if 'lat' in col.lower()), None)
lon_col = next((col for col in df.columns if 'long' in col.lower() or 'lon' in col.lower()), None)
site_col = next((col for col in df.columns if 'site' in col.lower()), None)

print(f"Detected Columns -> Latitude: '{lat_col}', Longitude: '{lon_col}', Site: '{site_col}'")

# 4. Loop through your dataframe safely using the detected columns
for index, record in df.iterrows():
    lat = record[lat_col]
    lon = record[lon_col]
    site_name = record[site_col]

    # Add a marker for each record
    folium.Marker(
        location=[lat, lon],
        icon=folium.Icon(color="red", icon="info-sign"),
        popup=f"Launch Site: {site_name}"
    ).add_to(marker_cluster)

# 5. Display the map
site_map
```

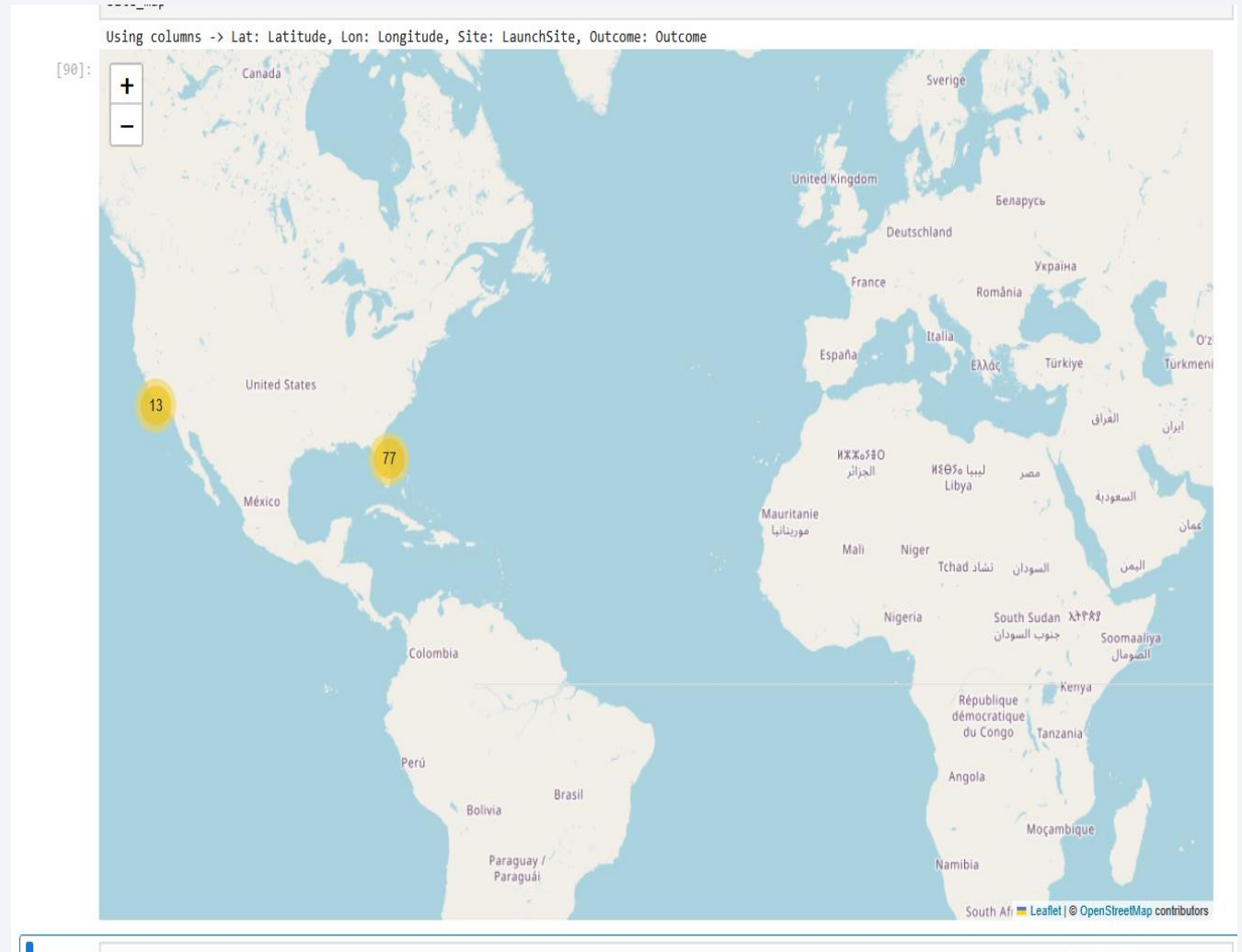


Launch Outcomes Visualization by Site

3. Explanation of Important Elements & Findings

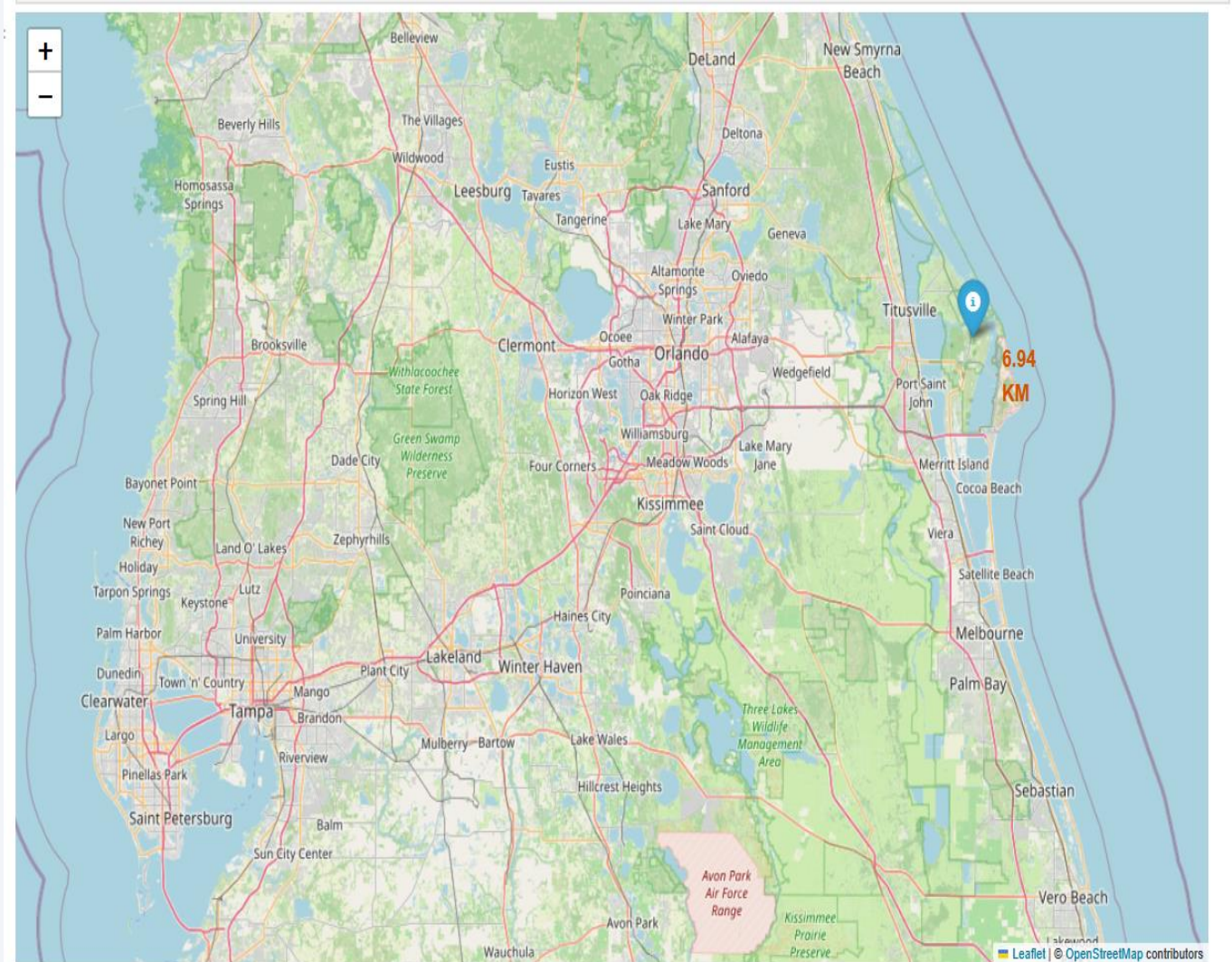
Include these details on your slide or presentation notes:

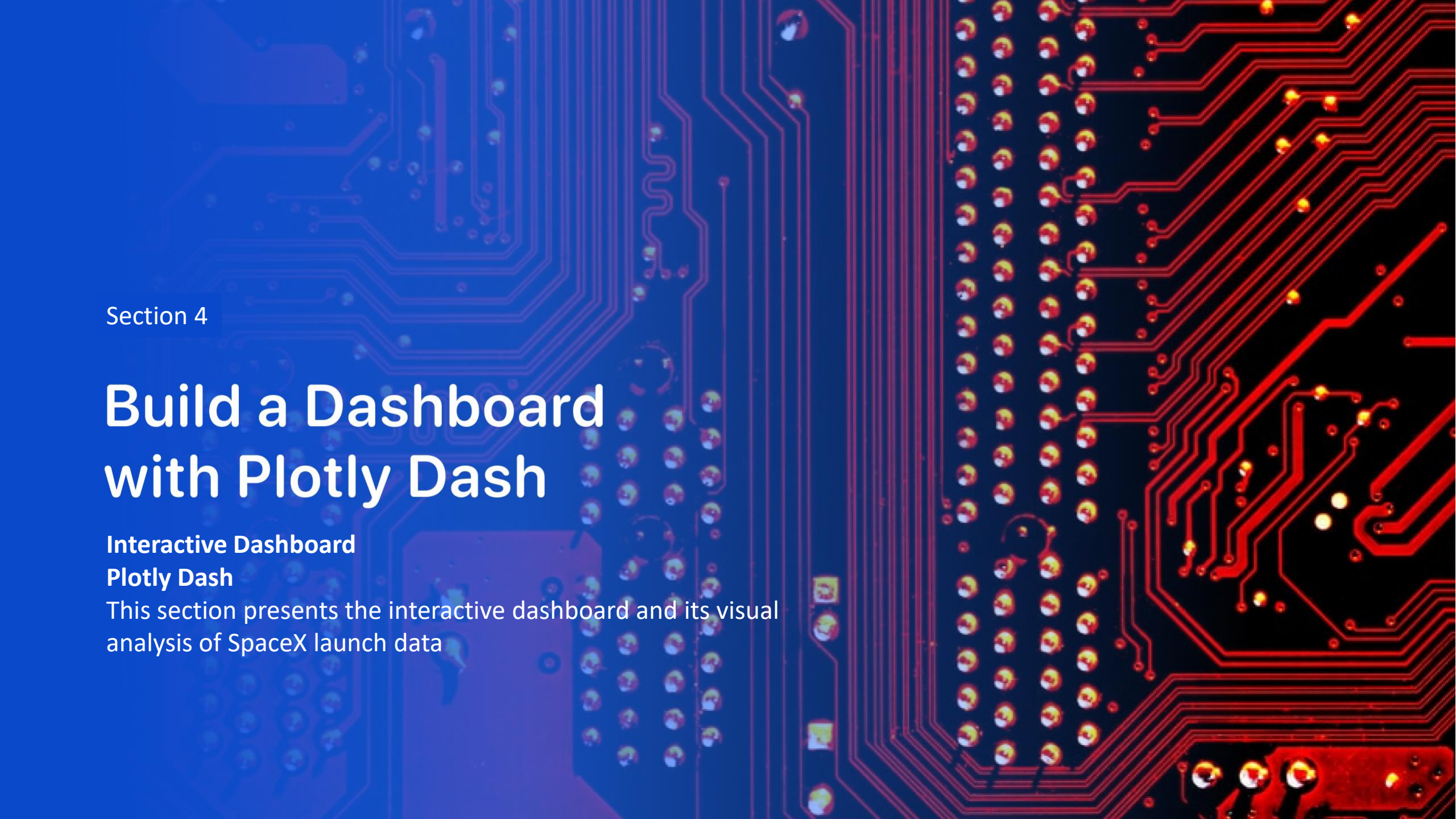
- **Color-Coded Markers:** Successes are marked distinctively (e.g., green markers) while failures or unsuccessful landings are highlighted in another color (e.g., red markers) for immediate visual distinction.
- **Interactive Clustering:** The map uses cluster icons displaying sub-counts when zoomed out, which expand into individual launch markers when zoomed in.
- **Key Findings:** Visual inspection reveals that certain launch pads achieved higher success concentrations, while early-stage or specific orbital trajectories experienced higher failure rates during experimental recovery attempts



Launch Site Proximity and Distance Analysis

- **3. Explanation of Important Elements & Findings**
- Include these points on your slide or presentation notes:
- **Proximity Lines:** PolyLines connect the launch site to nearby geographical or infrastructural landmarks (such as coastlines, highways, and railways) with calculated distances displayed directly on the map.
- **Geographical Advantage:** Most SpaceX launch facilities show close proximity to coastlines to allow safe rocket trajectories over water away from populated areas, while maintaining proximity to major highways and railways for logistics and transportation.





Section 4

Build a Dashboard with Plotly Dash

Interactive Dashboard
Plotly Dash

This section presents the interactive dashboard and its visual analysis of SpaceX launch data

Total Launch Success Breakdown Across All Sites

3. Explanation of Important Elements & Findings

Include these key points on your slide or presentation notes to address the findings requirement:

- **Pie Chart Representation:** The visualization uses proportional sectors to display the total distribution or success count of missions across different launch locations.
- **Site Comparison:** Highlights which launch facilities handled the highest volume of missions and contributed most significantly to successful orbital flights.
- **Key Findings:** Facilities like Kennedy Space Center and Cape Canaveral capture the largest share of successful launches due to their frequent usage, whereas sites with fewer total launches show lower aggregate success shares.

Total Launch Success by Site



Total Launch Success Breakdown Across All Sites

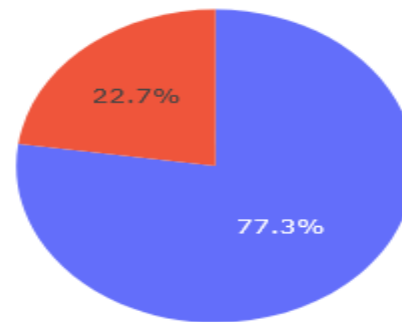
Why this approach works:

•**Dynamic Calculation**: Instead of hardcoding a specific site name, it programmatically computes the mean of the Class column grouped by LaunchSite to accurately identify the top-performing facility.

•**Targeted Visualization**: It filters the dataset specifically for that site so the pie chart clearly displays its individual success-to-failure proportions.

The launch site with the highest success ratio is: KSC LC 39A

Launch Success vs. Failure Breakdown for KSC LC 39A

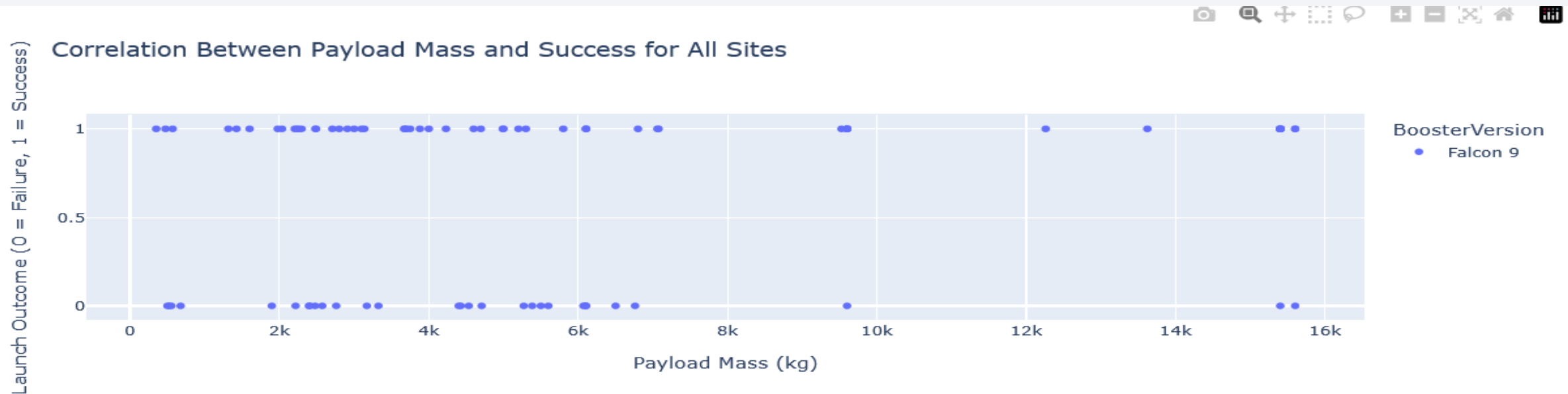


Total Launch Success Breakdown Across All Sites

3. Explanation of Important Elements & Findings

Include these key points on your slide or presentation notes :

- **Payload Range & Success Rates:** Mid-range payloads (typically between 2,000 kg and 9,500 kg) show a very high concentration of successful launch outcomes.
- **Booster Version Performance:** Certain advanced booster versions exhibit consistent success across varying payload weights, demonstrating enhanced reliability.
- **Heavy Payload Limitations:** Extremely heavy payloads (above 10,000 kg) present higher operational complexity, though successful booster recovery and orbital delivery rates improve significantly with newer rocket iterations.



Section 5

Predictive Analysis (Classification)

Classification Accuracy

- The pie chart presents the distribution of launch outcomes across launch locations.
- **Key Findings**
- The chart allows comparison of the contribution of different launch sites to the overall number of successful and unsuccessful missions.

The model with the highest classification accuracy is: Logistic Regression (0.833)

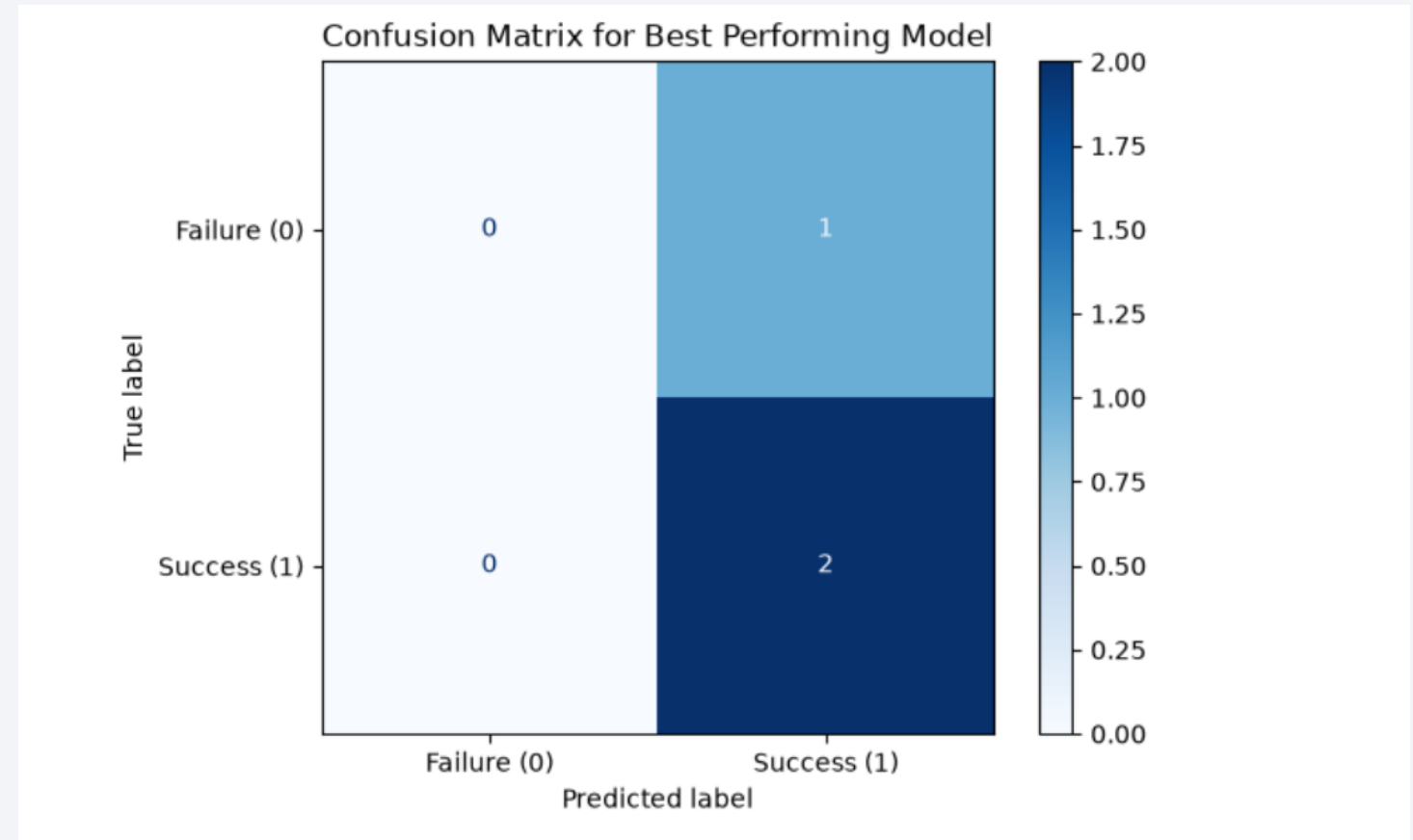


Classification Accuracy Comparison Across Models



Confusion Matrix

- The confusion matrix evaluates the predictions of the selected classification model by comparing actual landing outcomes with predicted outcomes.
- **Interpretation**
- **True Positive:** Successful landing correctly predicted.
- **True Negative:** Unsuccessful landing correctly predicted.
- **False Positive:** Unsuccessful landing predicted as successful.
- **False Negative:** Successful landing predicted as unsuccessful.



Conclusions

- **Launch Site:** The analysis shows differences in landing outcomes across launch sites.
- **Payload and Orbit:** Payload mass and orbit type show different distributions of landing outcomes.
- **Flight Experience:** Landing success patterns changed as the number of Falcon 9 flights increased.
- **Interactive Analysis:** Folium provided geographical insight into launch-site locations, while interactive visualizations supported exploration of the dataset.
- **Predictive Modeling:** Logistic Regression, SVM, KNN, and Decision Tree models were evaluated for predicting landing success.
- **Model Evaluation:** The final model comparison was based on test-set accuracy and confusion-matrix results.
- **Overall Project Finding**
 - The project demonstrates how data collection, data wrangling, exploratory analysis, SQL, visualization, and machine learning can be combined to analyze and predict Falcon 9 first-stage landing outcomes.

Appendix

- **Data Sources**
- SpaceX REST API
- Historical Falcon 9 launch records
- **Tools and Technologies**
- Python
- Pandas
- NumPy
- SQLite
- Seaborn
- Matplotlib
- Folium
- Plotly
- Plotly Dash
- Scikit-Learn
- GridSearchCV
- BeautifulSoup
- **GitHub Repository**
- <https://github.com/kooke12/IBM-Applied-Data-Science-Capstone.git>

Thank you!

